

Mobile Application Development (COMP2008)

Assignment 2 (Part B) – 15 Marks

Due Date: 26 October 2025, Time: 23:59 AWST

- Assignment 2 has two parts:
 - **Part A** – Build a **Pocket Library** App → **Group-based** (you may choose to work alone). Worth 25 marks. Due: **19th October**.
 - **Part B** – Reflection & Research → **Individual submission**. Worth 15 marks. Due: **26th October**.

Objective: The objective of this assignment is to reflect on the technical challenges encountered during the development of Android applications and provide a self-reflection on the learning process. This is an individual assignment requiring the submission of a well-formatted report. If you include source code and implementations, present them as snapshots within the report with clear explanations of how they address the challenges discussed. The focus is on critical analysis, practical problem-solving, and theoretical understanding of Android development. In addition, students are encouraged to identify and discuss recent research developments, emerging tools, and comparative techniques to evaluate how their approach aligns with or differs from current trends, making the reflection both a personal learning exercise and a brief survey of the state of the art.

Tasks (total 15 Marks):

Practical Reflection (3 Marks): Summarize your contributions to both Assignment 1 and Assignment 2. Focus on the key technical challenges encountered with Fragments, ViewModel, RecyclerView, and list management. Provide examples from your practical work, discussing how you addressed these challenges. Additionally, briefly mention any security concerns that arose during development, such as handling permissions or safeguarding user data.

Cross-App Development (3 Marks): Investigate how modern Android permission models and security mechanisms affect cross-app data sharing. Identify at least two real-world techniques or frameworks that allow apps to share data securely without compromising user consent, and critically evaluate their strengths, weaknesses, and potential improvements.

Compare Jetpack Compose's declarative UI approach with traditional Android XML-based UI design (3 Marks): Investigate and explain the major advantages and disadvantages of each approach. Based on your findings, recommend which type of organization would benefit most from adopting Jetpack Compose and which might prefer to continue with XML layouts, providing clear reasons for your recommendations.

Evaluating Kotlin and Jetpack Compose for Enterprise-Level Applications (3 Marks): Analyse Kotlin's advanced features such as null safety, extension functions,

and coroutines and their impact on code quality, security, and developer productivity in large-scale applications. Discuss the pros and cons of adopting Jetpack Compose in production environments, focusing on its performance, maintainability, and learning curve for teams transitioning from XML-based layouts. Evaluate security benefits, such as Kotlin's null safety reducing vulnerabilities, and assess the industry and community support for Kotlin and Jetpack Compose in terms of long-term security and viability for enterprise-level applications.

Long-running operations in Android and Secure Networking (3 Marks): Discuss best practices for handling long-running operations in Android, particularly network-related tasks. Explain why executing long-running operations on the main thread is risky, and how to mitigate these risks.

Marking Criteria for each task:

- **Excellent (3 Marks):** Clear, well-organized, and in-depth reflection or analysis. Demonstrates thorough understanding of technical concepts, and security implications, and provides real-world examples or comparisons.
- **Good (2 Marks):** Clear and mostly complete responses with good reflections or analysis. Demonstrates a solid understanding but may lack depth or specific examples.
- **Satisfactory (1 Mark):** Basic understanding of the topic with limited analysis. Some key points may be missing, and reflections may not be fully developed.
- **Unsatisfactory (0 Marks):** Incomplete or unclear responses. Lacks reflection, analysis, or understanding of the concepts, with no reference to technical challenges or security issues.

Submission Guidelines:

Submit your report in PDF format electronically via Turnitin before the specified deadline. Ensure that your report is original and adheres to academic integrity standards. The report should be well-formatted and clearly written, with all sources properly cited to avoid plagiarism. **It is an individual report, so make sure you don't share it with your group members or others. We don't accept high similarity scores in Turnitin, which may result in academic integrity issues.**

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Academic Integrity:

This is an assessable task. If you use someone else's work or obtain someone else's assistance to help complete the part of the assignment that is intended for you to complete yourself, you will have compromised the assessment. You will not receive marks for any parts of your submission that are not your original work. If you do not reference any external sources, you are committing plagiarism and collusion, and penalties for Academic Misconduct may apply. This version updates the submission guidelines to require PDF submission via Turnitin.